PROJECT DUNE / PLAYER + ADMIN EDITION

# DUNE

## User manual

Arrakis survival, spice, factions and the Guild network

DuneCore 1.0.5 | DuneTerrain 1.0.5 | DuneEss 1.0.1

Minecraft Java / Paper 1.21.4 / Java 21

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Fully documented Arrakis. Reserved chapters for four other planets.

HOW TO USE THIS BOOK

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Player guidance comes first, followed by command references and server administration. Each of the four reserved planet chapters contains its currently implemented profile and space for future additions. Configuration values shown here are shipped defaults; your administrator may change them.


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READ FIRST

# What is included

This manual describes Project Dune's current source and shipped defaults, including the Session 2 terrain, announcement and summon features and the Sniffer-model sandworm update. It covers DuneCore 1.0.5, DuneTerrain 1.0.5, DuneEss 1.0.1 and the updated resource pack. It is a guide to the implemented plugin, rather than a promise of future content.

| Component | What it provides |
| --- | --- |
| DuneCore | Factions, stillsuits, hostile sandworms, spice harvesting, sun exposure, water trading, politics, jobs, configurable menus and Guild sightings. |
| DuneEss | Solaris balances through Vault, spice sales, Guild travel, paged help, landmark teleports and admin bloom placement. |
| DuneTerrain | Planet world generation, safe spawn, a Guild navigator, Arrakis formations, deposits, blooms and cave sietches. |
| Dune Resource Pack | A desert Sniffer sandworm skin, haunted Fremen zombie skin and bright spice-gold suspicious sand. |

## Reading the defaults

Prices, spawn chances, timers and landscape spacing in this book are defaults. An administrator can change many of them. Spacing is a procedural generation setting, not a guarantee that every landmark lies exactly that many blocks from the next one. Value targets depend on available sandy terrain.

## What has been checked

The preceding release passed builds and automated tests. The Sniffer update adds entity-pair tests; the manual is reviewed against the final source. A live Paper/Velocity test is still needed to confirm movement, appearance and proxy delivery on your actual server. The four non-Arrakis chapters deliberately reserve space for future planet-specific gameplay; their existing world profiles are already implemented.

PLAYER GUIDE

# Your first session

## 1. Open the guide and choose a faction

Use `/dune help` for the paged chat guide, or `/duneess help menu` for its inventory version. `/dune` opens the Dune Navigator. A tagged Nether Star is supplied after joining; left- or right-click it to reopen the Navigator. `/dune menuitem` restores the item if you no longer have it.

Choose with `/dune choose fremen`, or substitute `harkonnen`, `corrino`, `atreides` or `ordos`. This records your faction, creates an economy account, supplies a kit and requests a free first arrival at your faction homeworld. A normal player cannot change an existing selection; an administrator with `dune.admin` can.

## 2. Equip the actual stillsuit

The kit is placed in your inventory, not automatically equipped. Wear all four tagged armor pieces. Ordinary leather armor, dyed armor or items renamed at an anvil are not substitutes. A complete stillsuit protects against the plugin's Arrakis heatstroke timer. Matching faction tags are also required for full-set bonuses.

## 3. Gather supplies and earn Solaris

Use the tagged netherite Spice Harvest Shovel on Arrakis suspicious sand. Pick up the Melange Spice drops and sell with `/duneess sell`. Cave sietch caches also contain tagged spice, food and building materials. Check your balance with `/duneess balance` before requesting paid Guild travel.

## A practical first expedition

Bring the shovel, a complete suit, food and a weapon. Locate shelter before taking on a worm. A default account begins at zero Solaris, so ordinary travel initially requires earnings or an admin grant. An unpaid CHOAM contract can be accepted without a fare, but you still need a route to Arrakis and time to meet its quota.

PLAYER GUIDE

# Menus and help

The Navigator is a configurable chest-style interface. Background colors reflect your faction. Buttons are actions, not free inventory items: clicking them opens a page or runs a player command. Menus expose the same systems available through chat.

| Navigator button | Action |
| --- | --- |
| Spacing Guild | Open the destination menu. |
| CHOAM Contracts | Browse the current hourly contracts and accept one. |
| Player Spice Exchange | Open shortcuts to post or cancel a listing, or leave a job. |
| Wallet / Sell Spice | Read the balance or sell tagged spice through DuneEss. |
| Stillsuit Kit | Receive another faction kit through /dune kit. |
| Winged Stillsuit Upgrade | Run the paid Elytra merge. |
| Choose Faction | Open the five playable faction selections. |

## Other ways to navigate

`/dune menu` and `/dune gui` also open the Navigator. `/dune exchange` opens the exchange menu; `/dune job` prints the actual listings and remaining quotas. Interacting with a tagged Spacing Guild Navigator opens DuneEss's travel menu. The automatically generated navigator is an invulnerable, stationary villager at planet spawn.

## Current menu details

The default DuneCore travel page displays only four destinations because its dynamic list is capped at four. Sigma Draconis IV is still available through `/duneess travel sigma_draconis_iv` or the DuneEss travel menu, which lists all five worlds. The exchange page has listing-management shortcuts but no dynamic listing browser; use `/dune job` to find a sponsor.

Help supports `/duneess help <page>`, `next`, `previous` and `menu`. The chat guide includes clickable navigation. Some introductory messages and older files omit Ordos, but it is a distinct playable faction in the current code.

PLAYER GUIDE

# Factions and homeworlds

| Faction | Homeworld | Implemented traits |
|---|---|---|
| Fremen | Arrakis | Two spice units per harvested deposit; brief Resistance I refreshes while moving on Arrakis. Kit has Unbreaking III and leggings with Swift Sneak III. |
| Harkonnen | Giedi Prime | Player-caused combat damage is multiplied by 1.15. A matching full stillsuit refreshes Strength II. |
| Corrino | Kaitain | Incoming combat damage from an entity is multiplied by 0.85. Kit armor has Protection III. This does not reduce every environmental damage type. |
| Atreides | Caladan | Movement events periodically refresh Speed I. A matching full stillsuit adds eight health points, equivalent to four hearts. |
| Ordos | Sigma Draconis IV | A matching suit enables five-second Shadow Step on main-hand Ender Pearl use. Kit includes a replenishing explosive wind charge. |

Faction passives use the recorded faction. Full-set effects additionally check that all four equipped pieces are tagged stillsuits belonging to your recorded faction. A mixed set can protect against sun exposure if every piece is tagged as a stillsuit, but does not grant a faction set bonus.

## Arrival and respawning

Faction selection requests travel to the corresponding homeworld without a normal paid fare. On a proxy network, this is a server-transfer request. On a single backend, the homeworld must already be loaded for a local arrival. Respawn is redirected to the homeworld only when that world is loaded on the current backend; there is no general cross-server respawn router.

## Choosing carefully

Fremen suits favor harvesting and sneaking; Harkonnen and Corrino favor combat; Atreides favor speed and extra full-set health; Ordos favor short concealment and explosives. Spacing Guild has appearance-related support but is not one of the five player-selectable factions.

PLAYER GUIDE

# Stillsuits, flight and Ordos equipment

## Armor that looks like leather but protects like netherite

Faction kits use colored leather stillsuits with custom armor attributes. Helmet, chest, legs and boots provide 3, 8, 6 and 3 armor points respectively. Each piece adds three armor toughness and 0.1 knockback resistance. Faction trim patterns distinguish the suits; ordinary armor does not acquire these properties by matching its color.

`/dune kit` requires a faction and adds another shovel and four armor pieces. The shovel is a tagged netherite shovel with Efficiency V and Unbreaking III. Kits do not include a general food allowance or a conventional melee weapon; sietch caches provide food, while weapons must be obtained separately.

## Winged stillsuit

To use `/dune elytra`, wear your faction's leather stillsuit chestplate, carry an Elytra in storage inventory and have 1,000,000 Solaris. The command consumes the carried Elytra, replaces the worn chestplate with a tagged winged Elytra and charges the price. The new item retains custom chest armor protection and faction identity, and qualifies as the stillsuit chest slot for heat protection and set bonuses. It is a purpose-built upgrade, not a general enchantment transfer service.

## Ordos Shadow Step and charge

With a matching full Ordos suit, using an Ender Pearl in the main hand grants five seconds of invisibility. The plugin temporarily stores and removes armor, then restores it, returning displaced pieces to inventory or dropping overflow. During removal you do not count as wearing a complete suit, including for sun protection.

The tagged Ordos Implosion Charge launches a wind charge and supplies a replacement one tick later. Its impact produces immediate primed TNT attributed to the shooter. Despite its name, it uses ordinary TNT explosion behavior; treat it as destructive explosives rather than a harmless movement tool.

PLAYER GUIDE

# Solaris, spice sales and Guild travel

Solaris is the currency supplied by DuneEss through Vault. The default spice price is 15 Solaris per unit, normal Guild fare is 100 and starting balance is zero. Balances can be negative by default, allowing contract fines to create debt. There are no player-to-player pay or general bank commands in this plugin's command set.

## Selling spice

`/duneess sell` sells all correctly tagged spice in storage inventory. It does not sell ordinary orange dye or an item renamed Melange Spice. With DuneCore enabled, a Spacing Trust levy is taken and unauthorized spice sales can create smuggling bounties. A sale of 100 spice has a gross value of 1,500 Solaris; the default 1.21% levy rounds up to 19, leaving 1,481 Solaris.

## Travelling

`/duneess travel` opens the five-world menu. `/duneess travel arrakis` requests a specific destination; use the internal world IDs listed in this book. If the destination is the local loaded world, you teleport to its spawn. If networking is enabled and the destination is remote, DuneEss asks the proxy to switch servers.

The appointed Spice Steward faction has free Arrakis travel. Other travelers normally pay the fare. A local failed teleport refunds the fare; a proxy request refunds only when sending the request itself fails. The proxy does not confirm destination arrival to this plugin, so an offline destination can still leave a fare charged.

## Arrival rights are a separate system

Paying the fare does not grant legal spice rights or necessarily prevent an arrival bounty. The current arrival rule treats the Spice Steward faction as sanctioned and uses shared PostgreSQL permits. Ordinary arrivals without a valid permit can be marked wanted when the database-backed arrival check is active. See the legal-rights chapter before organizing a multiplayer expedition.

ARRAKIS FIELD GUIDE

# The Arrakis landscape

Arrakis is fully implemented in this manual. Its world ID is `arrakis`. The terrain profile uses Desert biome, sand at the surface and red sand filler. Default base height is 70, height variation 11 and noise scale 0.02. It contains sparse ores, procedural dunes and Dune-specific landmarks, rather than a custom city map.

| Landmark or resource | Shipped generation setting |
| --- | --- |
| Cave sietches | About one anchor per 2,000-block spacing cell. |
| Untriggered spice blooms | About one anchor per 2,000-block spacing cell, where suitable sand is available. |
| Darkstone formations | 750-block spacing, 10-18-block radius, up to three plateaus. |
| Oasis trees | 300-block spacing; default trunk height seven. |
| Ordinary spice deposits | Two placement attempts per chunk. |
| Rich random spice patches | 35% chance per 600-block region; radius 12; target base value 5,000 Solaris. |

## The three-level formations

Darkstone formations use blackstone, basalt and polished blackstone with sand-topped flat terraces. Each formation deterministically chooses one to three nested levels. Height differences between levels are capped at five blocks. The configured default maximum rise of 14 can reduce intervals; a three-level formation can therefore use four-block steps rather than exceeding the total rise.

Trees have persistent leaves, and vanilla vegetation is disabled for Arrakis. The plugin places a protected Guild navigator at a safe spawn location. Sparse Arrakis ore attempts include coal, iron and copper between Y -48 and 48, with default vein size five.

## Existing terrain is preserved

Generation changes apply to newly generated chunks. Old chunks do not grow additional plateaus after an upgrade. Administrators must plan exploration into new terrain or a backed-up world regeneration if they want the new formations everywhere.

ARRAKIS FIELD GUIDE

# Harvesting melange

Suspicious sand on Arrakis is treated as a spice deposit. With the resource pack it appears bright spice-gold. Harvest it by breaking the block with the tagged Spice Harvest Shovel from your faction kit. A regular netherite shovel is rejected even if it has the same enchantments or name. Other worlds' suspicious sand does not use this Arrakis harvesting rule.

## Yield and provenance

Fremen harvest two tagged spice units per deposit; other factions normally get one. Outside an active player-sponsored job, the block drops Melange Spice, represented by tagged orange dye. Its lore records the miner, faction and the current unit price. Collect the drop before using `/duneess sell`.

Deposits earned from a player job are sold immediately and paid through the contract split, so they do not also drop a second spice item. If a Fremen's two-unit yield exceeds the final remaining contract quota by one, the extra unit can still drop as personal spice. Every successful harvest also advances an active CHOAM quota.

## Respawn behavior

The default regeneration delay is 120 seconds. At the end of that scheduled delay, the location becomes suspicious sand only if its chunk is loaded and the block is air. A replacement block prevents regeneration. There is no retry when the chunk is unloaded, and the scheduled delay is not a persistent offline timer across server restarts.

## Estimating value

Values advertised for a field are based on ordinary one-unit harvest at the configured spice price. Actual returns depend on how much sand was converted, how many deposits remain, faction yield, contract assignment and levies. A target field worth 15,000 Solaris at a 15-Solaris unit price seeks 1,000 deposits; it is not a guaranteed cash reward on discovery.

ARRAKIS FIELD GUIDE

# Spice blooms and spice blows

An untriggered generated spice bloom is marked by a 3-by-3 magma patch with a raised central magma block. The server announces its coordinates when the marker's chunk loads. Finding the marker is separate from activating its rich field.

## Triggering a blow

A player or other moving living entity touching the marker can trigger it. An arrow or trident impact at the marker also works. The trigger removes the lower magma patch, turns the raised center to glowstone and produces an explosion sound and particle. The source does not use a destructive world explosion for this bloom effect.

Nearby sandy surface blocks within the default 24-block radius become suspicious sand. The target is 15,000 Solaris of deposits, using DuneEss's spice price when available. Insufficient sand means fewer deposits. The announcement service identifies a bloom by its marker coordinates, so detonating a previously announced marker does not restart its existing fifteen-minute window.

## Admin placement

`/duneess admin spicebloom [player]` searches for a clear sandy patch about 50 blocks ahead of the selected online player. The player must be on the loaded Arrakis world. It places an untriggered magma marker and announces it. The command does not instantly harvest the field or place a worm.

## Two other types of rich field

Random suspicious-sand patches have their own procedural setting and a 5,000-Solaris default target. Defeating a sandworm creates a much larger field targeting 100,000 Solaris. The loaded-marker announcement is specifically for spice blooms, not an alert for every ordinary suspicious-sand deposit or random patch in the world.

ARRAKIS FIELD GUIDE

# Sietches and personal caches

Procedural cave sietches have a broad stone-arched entrance, sleeping and kitchen spaces, a winding descent, and a lower reservoir with a waterfall. Their default anchor spacing is 2,000 blocks. Landmarks are generated only where the terrain and region permit placement.

## What is in a cache

| Supply | Amount and properties |
| --- | --- |
| Tagged melange | 250 units: three stacks of 64 and one stack of 58. |
| Gold armor | Helmet, chestplate, leggings and boots with Protection II and Unbreaking II. |
| Oak planks | 32 blocks. |
| Bread | 16 items. |

Each marked cache is personal to a player. Right-clicking opens a separate cache inventory; taking another player's share does not exhaust yours. Clicking a cache item moves it into your storage inventory, leaving any portion that does not fit. Dragging into the cache is blocked.

## Closing the cache

Closing the cache automatically transfers its remaining contents to your inventory and drops any overflow at your location. The player is then marked as having claimed that sietch. You cannot repeatedly reopen the same cache for another reward. Make room before opening, and check the ground after closing if your inventory was full.

The cache's gold armor is useful loot, but it is not tagged as a stillsuit. Wearing it does not satisfy the Arrakis heat-protection or faction full-set requirements. Its spice is correctly tagged and can be sold, but claiming a chest is not harvesting deposits and does not advance CHOAM's harvested-spice counter.

At the default sale price, 250 cache spice has 3,750 Solaris gross value. With the normal Trust levy, the direct sale leaves 3,704 Solaris, before any separate consequences of unauthorized trade.

ARRAKIS FIELD GUIDE

# Sun exposure and shelter

Arrakis has a dedicated heatstroke timer. Default lethal exposure is 300 seconds: five minutes of uninterrupted qualifying sunlight. The plugin checks once each second and warns on the first exposed second and at thirty-second intervals.

## When the timer runs

All of these conditions must apply: you are alive on Arrakis, outside Creative and Spectator modes, not wearing a full tagged stillsuit, in clear weather, during daytime before world time 12300, and exposed at the top of the surface rather than beneath shelter.

The plugin shows a burning effect while exposure accumulates. Fire-tick damage caused by that active exposure effect is cancelled until the timer expires. At the lethal threshold, the plugin sets health to zero. Ordinary fire, lava and other hazards are separate; a stillsuit is not general fire immunity.

## What resets exposure

Taking cover, wearing a full tagged suit, reaching night or otherwise ceasing to meet the exposure conditions resets the timer. You do not need to wait for gradual recovery. The stored exposure entry is removed and the associated fire effect cleared.

## Arrakis defaults to persistent daytime

DuneTerrain disables the daylight cycle on Arrakis and sets world time to 6000 during setup. Under that default, waiting for a natural sunset is not a reliable survival plan. Use a suit or reach shelter. An operator can alter world time or game rules separately.

## Equipment edge cases

A winged stillsuit counts as the chest piece. A mixed-faction set can still satisfy heat protection if every equipped slot has the stillsuit tag, but faction set bonuses require matching tags. Ordos Shadow Step temporarily removes armor, so its five-second concealment also temporarily breaks the full stillsuit condition.

ARRAKIS FIELD GUIDE

# Shai-Hulud: model, AI and combat

The current sandworm uses a real Sniffer as the visible, damageable body. An invisible Husk supplies native zombie-family targeting, pursuit and melee attacks. The body follows its controller each server tick. It is a composite gameplay creature: a vanilla Sniffer silhouette with a desert worm skin, not a custom elongated sandworm mesh.

| Sandworm property | Default |
|---|---|
| Visible body | Adult Sniffer; scale 3.5; 500 health points. |
| Hostile controller | Invisible adult Husk; melee damage attribute 8; movement attribute 0.33. |
| Automatic population cap | Three visible sandworms in Arrakis. |
| Automatic successful-spawn interval | 3,600 seconds, saved in config. |
| Spawn chance | 35% per eligible player check when the cooldown permits. |
| Check frequency | Every ten seconds, after the initial scheduled delay. |

## Encounter rules

Natural spawns require Arrakis and a sandy surface near a player. The chosen location is about 22-40 blocks away; a worm already within the nearby-entity check can prevent another spawn for that player. The hour-long cooldown starts after a successful automatic spawn, not merely after an unsuccessful check.

The visible Sniffer can be attacked and killed. Damage to the hidden controller is cancelled, so it does not become a second reward-bearing enemy. The controller handles melee targeting; combat damage received depends on ordinary combat rules and applicable faction defenses. The worm's spawn produces cloud particles and a Warden roar.

## Persistence and conversion

Both parts carry saved partner UUIDs. Existing loaded Husk-based worms from 1.0.4 are converted to the visible Sniffer implementation without resetting their remaining health. The plugin repairs links after load and removes the counterpart when a linked entity dies. Live-server movement and collision behavior still need testing on the deployed Paper build.

ARRAKIS FIELD GUIDE

# Defeating and summoning a sandworm

## Death rewards

Killing the tagged visible body removes the invisible controller. The worm has no normal item drops or experience reward. Instead, it strikes a visual lightning effect and converts available sandy terrain around the death location into suspicious-sand deposits.

The field targets 100,000 Solaris at ordinary base yield. At 15 Solaris per unit this means a target of 6,667 deposits. Its radius is derived from the target, and each eligible position also has an 85% placement chance. Unsuitable terrain can reduce actual deposit count. The death message reports how much was actually placed; it does not deposit 100,000 Solaris into the killer's account.

If at least one deposit is placed, this new spice field receives an Arrakis coordinates warning and its own fifteen-minute Guild countdown. Harvest it using the same tagged shovel and contract rules as any other Arrakis deposit.

## Admin summon command

`/dune summon sandworm` summons at the executing player's current position on Arrakis. `/dune summon sandworm 120 76 -240` summons at explicit absolute X/Y/Z coordinates in the loaded Arrakis world. Console can use the coordinate form, without a leading slash in the console input.

Both forms require `dune.admin`, granted to operators by default. Coordinates must be finite numbers within the world border and build height, with open feet and head space. The coordinate-free form fails outside Arrakis. A server without a loaded Arrakis world cannot summon there.

Admin summons bypass the automatic population cap and do not alter the saved automatic cooldown. They use the same Sniffer body, zombie AI, configured attributes, sighting messages and death-field behavior as natural worms. Summoning can therefore deliberately create more than the default three-worm population.

ARRAKIS FIELD GUIDE

# Haunted Fremen and rich-field danger

Haunted Fremen are hostile zombie raiders, distinct from sandworms and from water-retaliation Angry Locals. They spawn on Arrakis near players and are more likely near high-value suspicious-sand fields. The resource pack gives zombies their haunted Fremen appearance.

| Raider property | Default |
| --- | --- |
| Population cap | 12 tagged raiders on Arrakis. |
| Regular chance per player check | 8%. |
| Chance near a rich field | 45%. |
| Rich-field search | 32-block radius; more than 5,000 Solaris of remaining base-value deposits. |
| Health / attack attribute | 36 health points / 6 attack damage. |
| Movement attribute | 0.28. |
| Gear | Iron sword and leather helmet, both with zero gear drop chance. |

## How a raider appears

Checks run every ten seconds. A nearby tagged raider within the local-entity check prevents another for that player. The spawner tries sandy positions roughly 16-32 blocks away with clear feet and head space, then targets the player. These are configured probability checks, not a promise that a raider appears after a fixed time.

Raider sunlight combustion is cancelled by the plugin. Their equipment is for behavior and appearance, not a dependable source of an iron sword. They may despawn when far away. Unlike sandworms, their death does not run the special worm spice-field reward.

## Value is based on the field still present

The rich-field scan counts nearby surface suspicious sand and multiplies by the spice price. At the default price, more than 5,000 Solaris requires at least 334 counted deposits. It does not inspect the total value of your inventory, sale proceeds or a hidden field ownership record. A shrinking harvested field can consequently stop qualifying for the increased chance.

ARRAKIS FIELD GUIDE

# The fifteen-minute Guild notice

Session 2 introduces a discovery window for sandworms and spice blooms. The first message is local to the Arrakis server, includes X/Y/Z and warns that the Spacing Guild will notify other worlds in fifteen minutes.

## Which events create sightings

- A generated, untriggered bloom marker is found when its chunk loads, including loaded chunks checked at server startup.
- An admin places a bloom marker; a previously unannounced marker erupts.
- A sandworm spawns or loads, including admin summons.
- A defeated sandworm produces a new field with at least one spice deposit.

## Countdown and global delivery

At the saved deadline, DuneCore sends a lore announcement through the proxy's standard BungeeCord Message channel to ALL online players. The Guild message includes the recorded sighting coordinates. With `network.enabled: false`, delivery is local instead.

The deadline uses elapsed real time, not a fifteen-minute Minecraft day. The service saves timers and identities in `plugins/DuneCore/arrakis-announcements.yml`. Restarting preserves the original deadline. Chunk reloads do not restart a countdown for the same bloom coordinates or worm UUID.

Plugin messaging requires an online player connection on the sending backend. If Arrakis is empty when the notice is due, it remains pending until a carrier is available. The standard channel has no delivery acknowledgment; the plugin records a notice as sent after submitting it, and cannot prove every proxy client received it.

## Interpreting coordinates

Coordinates are the initial sighting, not a live moving-worm tracker. A worm may move, despawn or die before the Guild message arrives. The notice also does not reserve the field for the discoverer, lock harvesting, or establish political ownership. The fifteen-minute window is an information embargo.

ARRAKIS FIELD GUIDE

# Finite water and selling buckets

Arrakis water is deliberately constrained. Natural source buckets collected through the water listener receive the Arrakis Water - Finite Source tag. `/dune water sell` sells these tagged buckets. Ordinary water buckets bought or collected elsewhere are not automatically accepted as Arrakis trade goods.

Water flow is cancelled on Arrakis. Emptying water buckets there is blocked, and the listener prevents new ice or snow formation and water-cauldron filling. The intended loop is to locate existing natural sources, collect finite buckets, and trade them rather than build a renewable placed-water farm. The source-tag check applies to natural water generally; it is not tied to ownership of one specific sietch.

## Price and legal status

Base price is half the spice price using integer division, with a minimum of one. At the default spice price of 15, base water value is seven Solaris per bucket. Members of the appointed House of Water receive the legal base price. Others are treated as smugglers and receive a 45% gross premium, rounded over the entire batch. The Trust levy is then taken from gross revenue.

For ten buckets, an authorized sale has gross value 70, levy one and net 69. An unauthorized sale has rounded gross value 102, levy two and net 100. Tagged buckets are removed by the sale; the command does not return empty buckets.

## Career commands versus actual authorization

`/dune water guild`, `smuggler` and `status` select or display an in-memory career label. The actual sale rule uses imperial House of Water authorization, not that chosen label. Choosing guild does not make an unappointed faction legal. The label resets on restart.

Unauthorized collection has a default 30% retaliation chance, potentially announcing a local bounty and spawning two Angry Local Husks targeting the collector. The collection retaliation listener currently also lacks an Arrakis-world guard; administrators should consider this when testing collection on other planets. Shared bounty records require PostgreSQL.

PLAYER SYSTEMS

# CHOAM hourly contracts

CHOAM jobs are server-issued quota contracts, separate from player-sponsored exchange jobs. `/dune serverjobs` lists the current offers. `/dune serverjobs accept small-spice-run` accepts a current template. The Navigator also has a CHOAM menu with an active-contract summary.

| Contract ID | Quota / pay / limit | Failure fine |
|---|---|---|
| small-spice-run | 100 spice / 3,000 Solaris / 30 min | 150 Solaris |
| guild-supply | 250 spice / 8,500 Solaris / 45 min | 425 Solaris |
| imperial-contract | 500 spice / 20,000 Solaris / 60 min | 1,000 Solaris |
| dangerous-harvest | 800 spice / 35,000 Solaris / 90 min | 1,750 Solaris |

Three shuffled offers are normally available each hour. The rotation follows UTC epoch hour boundaries, rather than counting sixty minutes from your last acceptance. Accepting is free, starts the template's own time limit and permits only one active CHOAM contract per player.

## Completing and failing

Successfully harvesting Arrakis deposits advances the quota by yielded units, including the Fremen two-unit yield. Selling carried spice or taking cache spice does not advance it. Reaching the quota pays the configured amount directly to the wallet. There is no requirement to hand over inventory spice to an NPC.

Failure costs 5% of the offered pay, rounded up. A ten-second check expires contracts even while a player is offline; real time continues through server downtime. With default negative balances enabled, a fine can create debt. An accepted contract retains its ID, progress and original acceptance time in local state. If its template is removed, it disappears without a template-based fine.

An administrator can edit server-jobs.yml and use `/dune serverjobs reload`. Offers remain limited to the current selection until the next rotation. Reloaded quota, pay and duration values are resolved from the current template, so changes can affect accepted contracts.

PLAYER SYSTEMS

# The player spice exchange

Player jobs split the sale value of newly harvested spice between a worker and sponsor. They are not prepaid procurement orders: the sponsor earns 85% of the harvest's base sale value and the worker earns 15%. Posting does not withdraw an escrow sum from the sponsor.

## Posting and accepting

`/dune job post 1000` creates your listing with a quota between one and 100,000. You may have one listing at a time. `/dune job` lists sponsors and remaining quota. `/dune job accept PlayerName` accepts someone else's open listing; a UUID is also accepted. You cannot take your own listing, take a listing already occupied or accept another while you are already a worker.

Accepting assigns the worker and requests `/duneess travel arrakis`. The assignment occurs before the travel request; a failed trip does not automatically cancel the assignment. You may need to fund the fare or contact the administrator before travelling manually.

## How payment works

Each credited harvested unit is paid immediately. Rounding is based on accumulated listing value, maintaining the 15/85 split across the job. At a 15-Solaris unit price, a completed 100-unit listing is worth 1,500: the worker receives 225 and sponsor 1,275. These direct contract payouts do not run the ordinary DuneEss direct-sale levy path.

Harvested job spice is not also dropped for a second sale. CHOAM progress is credited as well, so the two job systems can advance from the same successful harvesting action. Any yield above the remaining player-job quota falls back to personal spice.

## Leaving, cancellation and saving

`/dune job finish` leaves your worker assignment; it does not force-complete the quota. `/dune job cancel` cancels your own unclaimed listing, but refuses while a worker is assigned. Listings are local to a backend. Worker assignments are in memory and need reacceptance after restart. In shared-database mode, the current save path returns after saving player profiles and does not persist new exchange-listing changes, so do not rely on listing restart durability in that mode.

PLAYER SYSTEMS

# Politics, elections and the Spacing Trust

`/dune politics` and `/dune trust` access the same prototype political system. `/dune politics status` shows the Trust balance, election bucket, faction, leader, Emperor, stored appointments and invasion state. You need a selected faction for actions other than status.

## The treasury and elections

The default Trust levy is 1.21% of applicable gross direct sales, rounded up. `/dune politics deposit 500` voluntarily withdraws 500 from your wallet and adds it to the Trust if you have sufficient funds. The Trust is not a personal withdrawable bank account.

Use `/dune politics vote PlayerName` for your faction leader and `emperorvote PlayerName` for the Emperor. A candidate must be online and have a playable faction. Faction-leader candidates must belong to your house; imperial candidates may come from another house. Each player can cast one ballot in each applicable election bucket. The current highest tally determines the winner immediately, with lexicographic UUID tie-breaking.

Election buckets are seven-day Unix epoch divisions, not a scheduled election ceremony at a local Monday midnight. Leaders and the Emperor are recalculated from the current bucket's ballots. There is no separate close-election or campaign system.

## Invasions

The current elected faction leader can use `/dune politics invasion <house>` to target a different playable faction. Default cost is 50,000 from the Trust, duration sixty minutes and required defense contributions five. Only one unexpired invasion is allowed.

Members of the defending house contribute using `/dune politics defend`; each stored defender identity contributes once. Reaching five broadcasts that the invasion was repelled and clears its active deadline. This is a ledger-and-announcement prototype: there is no automatic territory capture, structure destruction, NPC army or attacker victory payout when time expires. Status can retain stored attacker and defender labels after expiry. Shared politics storage remains a whole-record prototype, not fully transactional multi-server governance.

PLAYER SYSTEMS

# Imperial rights, permits and bounties

## The Emperor's appointments

The current elected Emperor can appoint a Spice Steward with `/dune politics steward <house>` and a House of Water with `/dune politics houseofwater <house>`. Each office can be appointed once for the current Emperor and election bucket. The two offices may belong to different playable factions.

Rights depend on the current winning Emperor, matching stored term and appointed house. They can cease when the week changes or the winner changes. Status displays stored appointment labels, which may no longer be legally active. The Spice Steward faction has authorized spice trading and free Arrakis travel; the House of Water has authorized water trading. The present arrival exemption checks the Spice Steward, not both offices.

## Arrival permits and trade bounties

The sanctioned travel path can issue a two-minute, single-use PostgreSQL permit. On Arrakis arrival, a Steward member is exempt; otherwise a valid permit is consumed or the player is marked wanted. Without the shared database, enforcement logs its limitation rather than maintaining shared permits.

Unauthorized spice or water sales can issue a bounty equal to their gross sale value. Unauthorized arrival's default bounty is derived from half the spice price plus the 45% premium, yielding eleven Solaris with the default values. Water collection retaliation uses its own base-price calculation. These paths are related but not identical pricing formulas.

## Claiming a bounty

When another player kills a wanted player, the shared database claim pays the stored reward to the killer through DuneEss and marks the bounty claimed. Environmental deaths and self-kills do not claim it. Issuing a newer bounty overwrites the target's previous bounty record and reopens it; rewards do not accumulate as a stack of independent tickets.

Local witness announcements identify wanted players. Bounty storage and claims require PostgreSQL. There is no player bounty-board, pardon or fine-payment command in the current plugin. Lore about edicts in older documents does not imply an implemented player edict command.

COMMAND REFERENCE

# Everyday player commands

Angle brackets mean a required value; square brackets mean optional input. Do not type the brackets themselves. Player commands are generally executed in-game. World IDs use underscores, not the displayed planet names or proxy hyphens.

| Command | Purpose |
|---|---|
| /dune | Open the Navigator. |
| /dune help | Open the paged DuneEss guide in-game. |
| /dune menu / /dune gui | Alternative Navigator entry points. |
| /dune menuitem | Receive the tagged Nether Star navigator item. |
| /dune choose <faction> | Choose fremen, harkonnen, corrino, atreides or ordos. |
| /dune kit | Get another faction kit; requires a selected faction. |
| /dune elytra | Merge a carried Elytra with the worn stillsuit chestplate for 1,000,000 Solaris. |
| /duneess balance | View your Solaris balance. |
| /duneess sell | Sell all tagged inventory spice. |
| /duneess travel [planet] | Open destination selection or travel to a world ID. |
| /duneess help [page] | Show a help page; next, previous and prev are supported. |
| /duneess help menu | Open the inventory help guide. |
| /dune water status | Display the current career label and water-trade message. |
| /dune water guild / smuggler | Select a career label; actual legal pricing follows appointments. |
| /dune water sell | Sell all tagged Arrakis water buckets. |

Use DuneEss for balance, sales and travel. Typing `/dune balance`, `/dune sell`, `/dune travel` or `/dune admin` directly produces a relocation message; those chat aliases do not run the DuneEss action. Navigator buttons route the corresponding actions internally.

COMMAND REFERENCE

# Contracts and political commands

| Command | Purpose or condition |
| --- | --- |
| /dune serverjobs | List hourly CHOAM offers and your active progress. |
| /dune serverjobs accept <id> | Accept one currently offered template. |
| /dune exchange | Open exchange-management menu. |
| /dune job | Print player listings and remaining quotas. |
| /dune job post <quota> | Post one listing; quota 1-100,000. |
| /dune job accept <player> | Accept another player's free listing; name or UUID. |
| /dune job cancel | Cancel your own unclaimed listing. |
| /dune job finish | Leave your current worker assignment. |
| /dune politics status | View Trust, leaders, Emperor, appointments and invasion labels. |
| /dune politics deposit <amount> | Contribute positive affordable Solaris to the Trust. |
| /dune politics leader | Show your current elected faction leader. |
| /dune politics vote <player> | One same-faction leader ballot per bucket. |
| /dune politics emperorvote <player> | One Emperor ballot per bucket. |
| /dune politics invasion <house> | Elected leader only; charges the Trust. |
| /dune politics defend | Register a defending-house contribution. |
| /dune politics steward <house> | Current Emperor's one Spice Steward appointment. |
| /dune politics houseofwater <house> | Current Emperor's one House of Water appointment. |

`/dune trust` accepts the same politics subcommands. Political authority is earned from the current recorded vote tallies; being a server operator does not automatically make a player the elected leader or Emperor. The target house names are Fremen, Harkonnen, Corrino, Atreides and Ordos.

ADMIN GUIDE

# Admin command reference

All privileged actions listed here use the permission `dune.admin`, which defaults to operators. Grant it through the server's permission system only to trusted operators. Player context is required where the action depends on an inventory or current position.

| Command | Effect |
| --- | --- |
| /dune summon sandworm [x y z] | Summon a Sniffer-body worm on loaded Arrakis. Explicit coordinates work from console. |
| /dune guildnpc | Spawn a stationary tagged Guild villager at your current location. Unlike the generated spawn navigator, this command does not explicitly set invulnerability. |
| /dune givespice | Add one tagged spice item to your inventory. |
| /dune guireload | Reload DuneCore's gui.yml. |
| /dune serverjobs reload | Reload CHOAM templates; keep applicable current offers until next rotation. |
| /duneess balance [player] | Inspect an online player's balance. Console must name a player. |
| /duneess admin solari <player> <amount> | Grant a positive whole-number Solaris amount to an online player. |
| /duneess admin teleport arrakis [player] | Force teleport the target to loaded Arrakis spawn. |
| /duneess admin teleport sietch [player] | Search nearby generated sietch candidates and teleport to a found landmark. |
| /duneess admin teleport spiceblow [player] | Search for a generated or erupted bloom marker and teleport there. |
| /duneess admin spicebloom [player] | Place an untriggered bloom about 50 blocks ahead of an Arrakis player. |

Omit the optional player name to target yourself in-game. From console, supply an online player where required. Landmark teleport and bloom placement require the local Arrakis world; they do not generate a remote proxy world on demand.

An operator can use `/dune choose <faction>` to override an existing faction selection. There is no general live reload command for every gameplay config, no bounty-admin command and no command to regenerate all terrain. Restart after ordinary config edits unless a specific reload command is documented.

ADMIN GUIDE

# Installing and upgrading

The release targets Minecraft Java / Paper 1.21.4 and Java 21. Install Vault, DuneEss, DuneCore and DuneTerrain JARs in each Paper backend's plugins directory. The GitHub server package includes Vault 2.5.0; the manual companion package uses the same three project plugins and requires your existing Vault installation or that complete server package.

## Recommended upgrade sequence

- Stop the server cleanly and back up world folders, plugin folders and shared database state if used.
- Replace the previous JAR for each plugin with its new version. Keep only one active JAR per plugin; remove obsolete DuneRealms.jar when upgrading from that legacy plugin.
- Keep existing configuration and state files. DuneCore imports missing files from the legacy DuneRealms data folder; DuneEss can import legacy balances.
- Confirm each backend's selected planet and proxy mapping, then start the server and check enable messages.
- Replace the client resource pack, enable it and reload or reaccept it. A client using the old pack can display the new Sniffer with an old skin.

Existing worms are converted when loaded. The invisible controller is an implementation detail, not an extra player-visible enemy. Verify one admin summon and one natural encounter on a staging server before exposing the update to players.

## Resource pack installation

Vanilla and Fabric clients use the same ZIP in .minecraft/resourcepacks and enable it in Options > Resource Packs. It belongs in resourcepacks, not mods. Automatic delivery needs a direct-download URL configured as server.properties resource-pack; players must accept it. Texture changes alone do not change the Sniffer geometry.

## Build from source

Use Java 21 and Gradle 8.14 to run `gradle clean build` in each of the three module directories. This produces DuneCore 1.0.5, DuneTerrain 1.0.5 and DuneEss 1.0.1. Do not install the source ZIP as a plugin JAR.

ADMIN GUIDE

# World generation and primary-world setup

DuneTerrain selects one planet world per backend from `server.planet` in its config. Valid values are arrakis, giedi_prime, kaitain, caladan and sigma_draconis_iv. The world name matches that ID. All profiles remain in the file, but only the selected planet is created by the normal backend setup.

## Separate planet world

If Paper's primary world is the ordinary world named world, the plugin creates its selected separate planet after Paper loads its worlds and directly supplies its generator. It also chooses a safe spawn and optionally places the Guild navigator. New players who appear in the primary world are subsequently moved to the configured local planet spawn.

## Planet as Paper's primary world

If server.properties level-name is arrakis, Paper must know the generator during initial primary-world loading. In bukkit.yml, the worlds.arrakis section needs generator DuneTerrain:arrakis and biome-provider DuneTerrain:arrakis. For another selected planet, use that ID in all corresponding keys and values. Paper's global/world-default behavior files do not select this generator.

## Terrain changes and backups

The generator sets biome, height, filler, surface, water, vegetation and ores only during new chunk generation. Existing chunks are retained. If a world was generated with a different generator, changing config will not replace its rivers or hills. DuneTerrain logs a mismatch when it finds a selected world without its generator.

To intentionally regenerate a world, first stop and back up the server and move the specific old world folder aside before creating a fresh world. Never delete a live world's folder. Regeneration resets built structures and world-contained entities; player and plugin state needs separate consideration.

Arrakis setup freezes the daylight cycle at time 6000. Giedi Prime starts at 18000; other profiles start at 6000 with a running daylight cycle. Spawn X/Z defaults are zero for all five planets.

ADMIN GUIDE

# Gameplay tuning and menu configuration

| File | Settings and how they are used |
| --- | --- |
| plugins/DuneCore/config.yml | Sandworm and raider attributes, chances and limits; heatstroke; politics; water retaliation; proxy mapping; database. |
| plugins/DuneEss/config.yml | Spice price, fare, starting balance, negative-balance policy, shared settings inheritance. |
| plugins/DuneTerrain/config.yml | Selected planet, enabled structures, deposits and each planet's terrain profile. |
| plugins/DuneCore/server-jobs.yml | Offer count, template ID, title, quota, reward and time limit. Reload with the CHOAM admin command. |
| plugins/DuneCore/gui.yml | Menu size, title, backgrounds, icons, slots, labels, descriptions and allowed actions. Reload with /dune guireload. |

## What can be tuned

Natural worm frequency combines a successful-spawn cooldown, population cap, per-player chance and nearby-worm exclusion. Lowering one value does not guarantee a spawn if another condition prevents it. The config's runtime sandworms.next-spawn-at value is written after a natural spawn and retained across restart. Admin summons bypass that automatic gate.

Terrain max-plateaus and plateau-interval have hard caps of three and five even when a larger configured value is supplied. Other generation settings apply only to new chunks. Changing spice price changes future valuation and payout calculations; it does not create more deposits in a previously generated field.

## Customizing menus

Sizes must be multiples of nine between nine and 54; slot indices start at zero. Supported action types are open:<menu>, dune:<subcommand>, close and none. Dynamic lists support server_jobs, factions and planets. Placeholders include %player%, %balance%, %travel_cost%, %active_job%, and dynamic IDs/descriptions. Icons must be valid item materials; invalid items fall back to paper.

The default travel list can be raised from four to five items to include Sigma Draconis IV. Labels show the base fare, which may differ from a Steward's actual zero Arrakis fare. Existing water sietch-chunk-chance and edict-minutes settings do not by themselves activate the older proposed systems; use only behavior described in this manual.

ADMIN GUIDE

# The five-backend network

The intended topology is one Paper backend per planet behind a Velocity proxy. Each backend has the same project plugins, but a different selected local planet. DuneTerrain server.planet and DuneCore network.local-planet must match. DuneEss uses the DuneCore network settings by default.

| Displayed world | Plugin world ID | Default proxy server name |
|---|---|---|
| Arrakis | arrakis | arrakis |
| Giedi Prime | giedi_prime | giedi-prime |
| Kaitain | kaitain | kaitain |
| Caladan | caladan | caladan |
| Sigma Draconis IV | sigma_draconis_iv | sigma-draconis-iv |

The proxy server name must exactly match its configured name. Underscored world IDs and hyphenated proxy names are different strings. Guild travel uses the BungeeCord Connect channel. Global Guild sightings use Message with recipient ALL; the sending backend needs an online player connection.

## Proxy prerequisites

Velocity must have BungeeCord plugin-message compatibility enabled through bungee-plugin-message-channel. Configure matching forwarding on the proxy and each Paper backend, and keep backend ports protected from direct player connections. Follow your installed server version's official setup documentation for exact forwarding fields; the plugin is not a replacement for proxy authentication configuration.

## What is not a guaranteed network transfer

Connect is asynchronous with no success acknowledgment. The plugin cannot guarantee a destination is online or refund every failed remote arrival. Homeworld selection requests a transfer, but cross-server respawning needs separate routing. Job offers and exchange listings are backend-local. A GUI may show destinations that are not reachable until the administrator configures them.

Do not interpret the existence of all five terrain profiles as proof all five backend servers are running. The reserved planet pages describe the configured base worlds, not their deployment status.

ADMIN GUIDE

# Saving, shared data and recovery

| State | Current storage and behavior |
|---|---|
| Faction selection and inventory | Local faction data in players.yml by default; PostgreSQL player profiles when enabled. Shared inventory is saved on quit/profile saves and loaded on join. |
| Solaris | DuneEss balances.yml locally; optional shared dune_economy_balances PostgreSQL table. |
| Politics and Trust | politics.yml locally or a shared YAML record in dune_politics. |
| CHOAM offers and accepted jobs | Local server-jobs-state.yml, with original acceptance timestamps. |
| Player exchange | Local listings in players.yml in local mode; workers in memory. Shared-mode listing persistence is incomplete. |
| Guild sightings | Local arrakis-announcements.yml with deadlines and duplicate identities. |
| Permits and bounties | PostgreSQL tables; no equivalent shared local-file implementation. |
| Sietch cache claims | Player persistent tags for each cache ID. These are not included as a separate shared profile field. |

## Enabling PostgreSQL

DuneCore config has database.enabled, jdbc-url, username and password. It creates its profile, politics, permit and bounty tables when initialized. If required shared database initialization fails, DuneCore disables rather than starting its gameplay without that connection. DuneEss normally reuses DuneCore's database settings when enabled.

DuneEss maintains a separate shared economy table and can import legacy database or local balances when that table is empty. A shared-economy initialization failure can fall back to its local ledger; inspect logs rather than assuming the whole network still uses one authoritative balance store.

## Scope of synchronization

The current profile blob contains inventory and faction. It does not synchronize every aspect of Minecraft player state, every plugin tag or the local job files. Whole-record politics saving and multi-step payments are not fully transactional across simultaneous backends. Back up the database and local files together before changing storage mode. Do not edit runtime state while the plugins are active.

SUPPORT

# Troubleshooting and current limits

| Symptom | What to check |
| --- | --- |
| Sandworm still looks like a Husk | Confirm DuneCore 1.0.5 is the only installed Core JAR and the world has loaded the converted worm. A skin alone cannot change a Husk into a Sniffer. |
| Sniffer looks like ordinary green Sniffer | Replace and enable the updated resource pack; check automatic download caching and client acceptance. |
| Cannot summon | Check dune.admin, loaded Arrakis, numeric coordinates, border/build-height and open space. From console include x y z. |
| Cannot harvest a deposit | Use the tagged kit shovel, not ordinary netherite or an anvil rename, and ensure you are on Arrakis. |
| Burning or heatstroke | Equip all four tagged suit slots or take shelter. Default Arrakis time does not progress into night. |
| No Guild global notice at deadline | Check pending state, a player on Arrakis, network.enabled and proxy channel compatibility. Restart does not reset the deadline. |
| Travel unavailable or fare lost | Check proxy names and destination availability; Connect has no arrival acknowledgment. |
| No new plateaus near spawn | Existing chunks retain their earlier generation. Explore new chunks or arrange backed-up regeneration. |

## Behavior to understand before running a public economy

Water career labels do not determine legal pricing. Smuggling and arrival authorization require correct political state and PostgreSQL for shared enforcement. Operator status is not a substitute for an elected Emperor. CHOAM deadlines run through offline time, and fines can create debt. Ordinary deposit respawn delays do not persist through a restart.

The plugin has no dedicated thirst meter, custom worm mesh, built-in cities, territory-capture engine, general bank/pay system or universal multiplayer inventory transaction manager. The Sniffer body follows a hidden vanilla controller, so its collision and appearance should be checked in a live staging environment. These are current implementation boundaries, not features implied by the Dune lore names.

RESERVED PLANET CHAPTER

# Giedi Prime

This is a placeholder chapter for future Giedi Prime gameplay documentation. It is deliberately reserved; Arrakis has its own complete guide earlier in this book. The values below describe the base terrain profile that already exists, not planned content presented as released features.

| Current profile | Implemented default |
| --- | --- |
| Faction / world ID | Harkonnen / giedi_prime. |
| Terrain | Swamp biome, blackstone surface, deepslate filler. |
| Height | Base 61; variation five; noise scale 0.02. |
| Water | Level 63; swamp pools enabled; coverage 0.18. |
| Vegetation | No generated trees or flowers. |
| Ores | Broad coal, iron, copper, gold, redstone, lapis, diamond and emerald generation. Default Y -48 to 48. |
| Time and proxy | Initial time 18000; proxy name giedi-prime. |

## Reserved additions

Future documentation can describe industrial landmarks, routes, special resources, hazards and planet-specific objectives when those are implemented. Harkonnen abilities, Guild travel and shared systems already follow the relevant general chapters.

## Planet notes and future expansion

RESERVED PLANET CHAPTER

# Kaitain

This is a placeholder chapter for future Kaitain gameplay documentation. The current world profile is available, but this page does not invent imperial cities, missions or additional mechanics.

| Current profile | Implemented default |
| --- | --- |
| Faction / world ID | Corrino / kaitain. |
| Terrain | Plains biome, grass surface, dirt filler. |
| Height | Base 70; variation three; noise scale 0.02. |
| Water | Level zero; swamp pools disabled. |
| Vegetation | No generated trees; 64 flower placement attempts per chunk. |
| Ores | Coal, iron, gold, diamond and emerald; notably high gold and diamond attempt counts. Default Y -48 to 48. |
| Time and proxy | Initial time 6000; proxy name kaitain. |

## Reserved additions

Reserve this space for implemented imperial landmarks, economic activities, regional hazards and routes. Corrino armor and combat passives are described in the faction chapter; political elections and appointments are general systems rather than a Kaitain-only quest line.

## Planet notes and future expansion

RESERVED PLANET CHAPTER

# Caladan

This is a placeholder chapter for future Caladan gameplay documentation. The terrain and travel destination already exist; the later expanded chapter can document released local activities without changing the general player guide.

| Current profile | Implemented default |
| --- | --- |
| Faction / world ID | Atreides / caladan. |
| Terrain | Jungle biome, grass surface, dirt filler. |
| Height | Base 67; variation ten; noise scale 0.02. |
| Water | Level 72; pools enabled; coverage 0.55. |
| Vegetation | Trees enabled with 0.035 chance; no flower placement attempts. |
| Ores | Coal, iron, copper and diamonds; default Y -48 to 48. |
| Time and proxy | Initial time 6000; proxy name caladan. |

## Reserved additions

Future entries can cover coastlines, routes, settlements and water-world activities when implemented. Caladan's profile is a procedural jungle-and-water world; it is not a hand-built Atreides capital. The current Atreides Speed and full-set health bonuses work through the faction system.

## Planet notes and future expansion

RESERVED PLANET CHAPTER

# Sigma Draconis IV

This is a placeholder chapter for future Sigma Draconis IV gameplay documentation. The fifth planet is already a valid world ID and Guild destination, even though the default DuneCore travel GUI displays only four entries.

| Current profile | Implemented default |
| --- | --- |
| Faction / world ID | Ordos / sigma_draconis_iv. |
| Terrain | Snowy Plains biome, snow-block surface, deepslate filler. |
| Height | Base 72; variation 14; noise scale 0.02. |
| Water | Level zero; swamp pools disabled. |
| Vegetation | No generated trees or flower attempts. |
| Ores | Coal, iron, copper and diamonds; default Y -48 to 48. |
| Time and proxy | Initial time 6000; proxy name sigma-draconis-iv. |

## Reserved additions

Reserve future space for cold-world landmarks, exploration and Ordos-specific objectives that are actually released. Snowy terrain does not currently introduce a custom freezing meter or survival system. Ordos's implemented Shadow Step and explosive charge are described earlier.

Use `/duneess travel sigma_draconis_iv` or the DuneEss travel menu to reach this configured destination; operators can increase the Navigator's dynamic max-items to five.

## Planet notes and future expansion

REFERENCE

# Glossary and source notes

| Term | Meaning in this release |
| --- | --- |
| Tagged item | An item carrying the plugin's persistent identifier, rather than only a display name. |
| Spice bloom / blow | A magma-marked discovery; its triggered eruption converts nearby sand into spice. |
| Base field value | Deposit count multiplied by spice price at ordinary one-unit yield; not a guaranteed cash payout. |
| CHOAM contract | Hourly server-issued quota job with a time limit, reward and failure fine. |
| Player exchange job | Sponsor listing whose harvested value is split 15% worker and 85% sponsor. |
| Steward / House of Water | Current imperial appointments granting legal commodity rights. |
| Backend / proxy | A planet's Paper server / the Velocity server routing players between backends. |
| Sniffer body / Husk controller | The visible, damageable sandworm / its invisible native zombie AI partner. |

## Source of truth

This guide was checked against the source in unimatrixgit/mcdune, including DuneCore.java, SandwormMobs.java, ArrakisAnnouncements.java, ArrakisWater.java, Politics.java, ServerJobs.java, SharedDatabase.java, ConfigurableGui.java, DuneEss.java, DuneEconomy.java, DuneTerrain.java and RockPlateaus.java. Shipped config.yml, gui.yml, server-jobs.yml and plugin.yml files supply default settings and versions. Code takes precedence over conflicting older README text.

Session 2 originally shipped in commit 2e81c9d. The companion update uses DuneCore 1.0.5 for the real Sniffer body and zombie controller, plus the updated resource pack. The admin summon command is already part of the plugin; this book is not an instruction to add it manually.

## Texture production note

The new Sniffer skin was generated with the built-in image-generation tool using a prompt for a desert-tan, ochre and sand-brown segmented hide, ivory toothlike head details, sharply pixelated Minecraft texture, exact Sniffer UV placement and transparent unused areas. It was fitted to the native 192x192 atlas and original alpha mask. Model geometry remains the vanilla Sniffer. The following appendix reproduces every shipped config.yml leaf value for reference.

ADMIN APPENDIX

# Default configuration reference

Every setting below is copied from the release configuration files. Existing server files can differ because upgrades keep their saved configuration. Paths are relative to the corresponding plugin config.yml. Full GUI layouts are controlled separately in gui.yml; contract templates are in server-jobs.yml.

## DuneCore

| Setting | Shipped default |
|---|---|
| spice-respawn-seconds | 120 |
| sandworms.enabled | true |
| sandworms.health | 500.0 |
| sandworms.attack-damage | 8.0 |
| sandworms.movement-speed | 0.33 |
| sandworms.scale | 3.5 |
| sandworms.max-per-world | 3 |
| sandworms.spawn-chance | 0.35 |
| sandworms.spawn-interval-seconds | 3600 |
| spice-raiders.enabled | true |
| spice-raiders.max-per-world | 12 |
| spice-raiders.chance-per-player-check | 0.08 |
| spice-raiders.chance-near-rich-field | 0.45 |
| spice-raiders.field-radius | 32 |
| spice-raiders.field-value-threshold | 5000 |
| spice-raiders.health | 36.0 |
| spice-raiders.attack-damage | 6.0 |
| spice-raiders.movement-speed | 0.28 |
| arrakis-sun.enabled | true |
| arrakis-sun.lethal-seconds | 300 |
| arrakis-sun.warning-interval-seconds | 30 |

| Setting | Shipped default |
|---|---|
| network.enabled | true |
| network.local-planet | arrakis |
| network.servers.arrakis | arrakis |
| network.servers.giedi_prime | giedi-prime |
| network.servers.kaitain | kaitain |
| network.servers.caladan | caladan |
| network.servers.sigma_draconis_iv | sigma-draconis-iv |
| politics.trust-levy | 0.0121 |
| politics.invasion-cost | 50000 |
| politics.invasion-minutes | 60 |
| politics.defenders-needed | 5 |
| politics.arrakis-edict-minutes | 60 |
| arrakis-water.smuggler-retaliation-chance | 0.3 |
| arrakis-water.sietch-chunk-chance | 90 |
| database.enabled | false |
| database.jdbc-url | jdbc:postgresql://127.0.0.1:5432/dunerealms |
| database.username | dune |
| database.password | CHANGE_ME |

## DuneEss

| Setting | Shipped default |
|---|---|
| currency.singular | Solaris |
| currency.plural | Solaris |
| currency.spice-price | 15 |
| currency.travel-price | 100 |
| currency.starting-balance | 0 |
| currency.allow-negative-balances | true |
| database.use-core-database | true |
| network.use-core-settings | true |
| network.enabled | true |
| network.local-planet | arrakis |
| worlds | ["arrakis", "giedi_prime", "kaitain", "caladan", "sigma_draconis_iv"] |

## DuneTerrain

| Setting | Shipped default |
|---|---|
| server.planet | arrakis |
| guild-npc.enabled | true |
| sietches.enabled | true |
| sietches.spacing | 2000 |
| spice-blow.enabled | true |
| spice-blow.spacing | 2000 |
| spice-blow.radius | 24 |
| spice-blow.target-value | 15000 |
| spice-blow.unit-price | 15 |
| spice-deposits.per-chunk | 2 |
| spice-deposits.patches.enabled | true |
| spice-deposits.patches.region-spacing | 600 |
| spice-deposits.patches.chance-per-region | 0.35 |
| spice-deposits.patches.radius | 12 |
| spice-deposits.patches.target-value | 5000 |
| spice-deposits.patches.unit-price | 15 |
| planets.arrakis.enabled | true |
| planets.arrakis.spawn.x | 0 |
| planets.arrakis.spawn.z | 0 |
| planets.arrakis.terrain.base-height | 70 |
| planets.arrakis.terrain.height-variation | 11 |
| planets.arrakis.terrain.noise-scale | 0.02 |
| planets.arrakis.biome | DESERT |
| planets.arrakis.surface | SAND |
| planets.arrakis.filler | RED_SAND |

| Setting | Shipped default |
|---|---|
| planets.arrakis.water.level | 63 |
| planets.arrakis.water.swamp-pools | false |
| planets.arrakis.water.coverage | 0.18 |
| planets.arrakis.vegetation.trees | false |
| planets.arrakis.vegetation.tree-chance | 0.0 |
| planets.arrakis.vegetation.flowers-per-chunk | 0 |
| planets.arrakis.oasis-trees.enabled | true |
| planets.arrakis.oasis-trees.spacing | 300 |
| planets.arrakis.oasis-trees.trunk-height | 7 |
| planets.arrakis.darkstone-islands.enabled | true |
| planets.arrakis.darkstone-islands.spacing | 750 |
| planets.arrakis.darkstone-islands.min-radius | 10 |
| planets.arrakis.darkstone-islands.max-radius | 18 |
| planets.arrakis.darkstone-islands.max-height | 14 |
| planets.arrakis.darkstone-islands.max-plateaus | 3 |
| planets.arrakis.darkstone-islands.plateau-interval | 5 |
| planets.arrakis.ores.min-y | -48 |
| planets.arrakis.ores.max-y | 48 |
| planets.arrakis.ores.vein-size | 5 |
| planets.arrakis.ores.COAL_ORE | 2 |
| planets.arrakis.ores.IRON_ORE | 2 |
| planets.arrakis.ores.COPPER_ORE | 1 |
| planets.giedi_prime.enabled | true |
| planets.giedi_prime.spawn.x | 0 |
| planets.giedi_prime.spawn.z | 0 |
| planets.giedi_prime.terrain.base-height | 61 |

| Setting | Shipped default |
|---|---|
| planets.giedi_prime.terrain.height-variation | 5 |
| planets.giedi_prime.terrain.noise-scale | 0.02 |
| planets.giedi_prime.biome | SWAMP |
| planets.giedi_prime.surface | BLACKSTONE |
| planets.giedi_prime.filler | DEEPSLATE |
| planets.giedi_prime.water.level | 63 |
| planets.giedi_prime.water.swamp-pools | true |
| planets.giedi_prime.water.coverage | 0.18 |
| planets.giedi_prime.vegetation.trees | false |
| planets.giedi_prime.vegetation.tree-chance | 0.0 |
| planets.giedi_prime.vegetation.flowers-per-chunk | 0 |
| planets.giedi_prime.ores.min-y | -48 |
| planets.giedi_prime.ores.max-y | 48 |
| planets.giedi_prime.ores.vein-size | 5 |
| planets.giedi_prime.ores.COAL_ORE | 14 |
| planets.giedi_prime.ores.IRON_ORE | 12 |
| planets.giedi_prime.ores.COPPER_ORE | 9 |
| planets.giedi_prime.ores.GOLD_ORE | 7 |
| planets.giedi_prime.ores.REDSTONE_ORE | 8 |
| planets.giedi_prime.ores.LAPIS_ORE | 5 |
| planets.giedi_prime.ores.DIAMOND_ORE | 4 |
| planets.giedi_prime.ores.EMERALD_ORE | 2 |
| planets.kaitain.enabled | true |
| planets.kaitain.spawn.x | 0 |
| planets.kaitain.spawn.z | 0 |
| planets.kaitain.terrain.base-height | 70 |

| Setting | Shipped default |
|---|---|
| planets.kaitain.terrain.height-variation | 3 |
| planets.kaitain.terrain.noise-scale | 0.02 |
| planets.kaitain.biome | PLAINS |
| planets.kaitain.surface | GRASS_BLOCK |
| planets.kaitain.filler | DIRT |
| planets.kaitain.water.level | 0 |
| planets.kaitain.water.swamp-pools | false |
| planets.kaitain.water.coverage | 0.18 |
| planets.kaitain.vegetation.trees | false |
| planets.kaitain.vegetation.tree-chance | 0.0 |
| planets.kaitain.vegetation.flowers-per-chunk | 64 |
| planets.kaitain.ores.min-y | -48 |
| planets.kaitain.ores.max-y | 48 |
| planets.kaitain.ores.vein-size | 5 |
| planets.kaitain.ores.COAL_ORE | 4 |
| planets.kaitain.ores.IRON_ORE | 5 |
| planets.kaitain.ores.GOLD_ORE | 15 |
| planets.kaitain.ores.DIAMOND_ORE | 18 |
| planets.kaitain.ores.EMERALD_ORE | 3 |
| planets.caladan.enabled | true |
| planets.caladan.spawn.x | 0 |
| planets.caladan.spawn.z | 0 |
| planets.caladan.terrain.base-height | 67 |
| planets.caladan.terrain.height-variation | 10 |
| planets.caladan.terrain.noise-scale | 0.02 |
| planets.caladan.biome | JUNGLE |

| Setting | Shipped default |
| --- | --- |
| planets.caladan.surface | GRASS_BLOCK |
| planets.caladan.filler | DIRT |
| planets.caladan.water.level | 72 |
| planets.caladan.water.swamp-pools | true |
| planets.caladan.water.coverage | 0.55 |
| planets.caladan.vegetation.trees | true |
| planets.caladan.vegetation.tree-chance | 0.035 |
| planets.caladan.vegetation.flowers-per-chunk | 0 |
| planets.caladan.ores.min-y | -48 |
| planets.caladan.ores.max-y | 48 |
| planets.caladan.ores.vein-size | 5 |
| planets.caladan.ores.COAL_ORE | 4 |
| planets.caladan.ores.IRON_ORE | 5 |
| planets.caladan.ores.COPPER_ORE | 5 |
| planets.caladan.ores.DIAMOND_ORE | 2 |
| planets.sigma_draconis_iv.enabled | true |
| planets.sigma_draconis_iv.spawn.x | 0 |
| planets.sigma_draconis_iv.spawn.z | 0 |
| planets.sigma_draconis_iv.terrain.base-height | 72 |
| planets.sigma_draconis_iv.terrain.height-variation | 14 |
| planets.sigma_draconis_iv.terrain.noise-scale | 0.02 |
| planets.sigma_draconis_iv.biome | SNOWY_PLAINS |
| planets.sigma_draconis_iv.surface | SNOW_BLOCK |
| planets.sigma_draconis_iv.filler | DEEPSLATE |
| planets.sigma_draconis_iv.water.level | 0 |
| planets.sigma_draconis_iv.water.swamp-pools | false |

| Setting | Shipped default |
| --- | --- |
| planets.sigma_draconis_iv.water.coverage | 0.18 |
| planets.sigma_draconis_iv.vegetation.trees | false |
| planets.sigma_draconis_iv.vegetation.tree-chance | 0.0 |
| planets.sigma_draconis_iv.vegetation.flowers-per-chunk | 0 |
| planets.sigma_draconis_iv.ores.min-y | -48 |
| planets.sigma_draconis_iv.ores.max-y | 48 |
| planets.sigma_draconis_iv.ores.vein-size | 5 |
| planets.sigma_draconis_iv.ores.COAL_ORE | 5 |
| planets.sigma_draconis_iv.ores.IRON_ORE | 5 |
| planets.sigma_draconis_iv.ores.COPPER_ORE | 2 |
| planets.sigma_draconis_iv.ores.DIAMOND_ORE | 2 |